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MATHEMATICS
ESSAY PROBLEMS

Name : _____ **Index Number** : _____

Country : _____

17th International Mathematics and Science Olympiad

Indonesia

21 January 2021

Instructions:

1. Write your name, country and index number on every page of this booklet.
2. Write your detailed solution in Arabic Numerical or English in the space provided for each question in this booklet. If you need more space for your working solutions, you may use the reverse side of each page, but please write "next side continued" in the last line.
3. There are 13 questions in this paper.
4. Each question is worth 3 marks and partial credits may be awarded.
5. Diagrams are NOT drawn to scale. They are intended only as aids.
6. You have 90 minutes to complete this paper.
7. Use black pen or blue pen or pencil to write your answer.

The following table is for jury use only.

No.	1	2	3	4	5	6	7	8	9	10	11	12	13	Total
Score														
Signature														
Score														
Signature														

Do not turn over this page until you are told to do so.

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

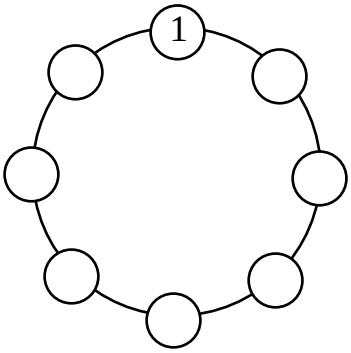
1. Mali tries to insert some number of " + " signs between the digits of 777777777777 (twelve copies of 7s) so that the resulting expression is a multiple of 30. How many ways can she do it?

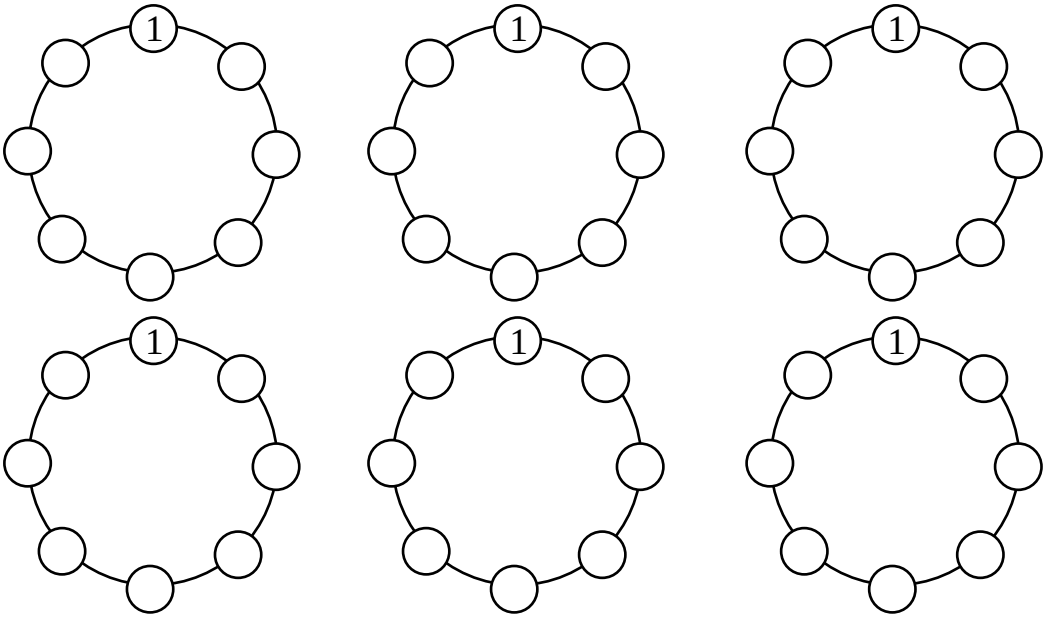
Answer : _____ ways

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

2. Place each of the numbers 2, 3, 4, 5, 6, 7 and 8 exactly once in the eight empty circles (the position of 1 is already fixed) such that the sum of each pair of adjacent numbers is a prime number. Note that the diagrams are considered the same if it can be obtained by reflection or rotation. List down all possible diagrams.





Answer : _____

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

3. Four runners (A, B, C and D) participated in a race. Before the event, all four of them (I, II, III and IV) anonymously predicted the final rankings as shown in the table below:

	1st	2nd	3rd	4th
I's prediction	B	C	A	D
II's prediction	C	A	D	B
III's prediction	D	B	A	C
IV's prediction	B	C	D	A

If I and III got 2 correct predictions each while II and IV got 1 correct prediction each. Determine the final placings of the four participants.

Answer :

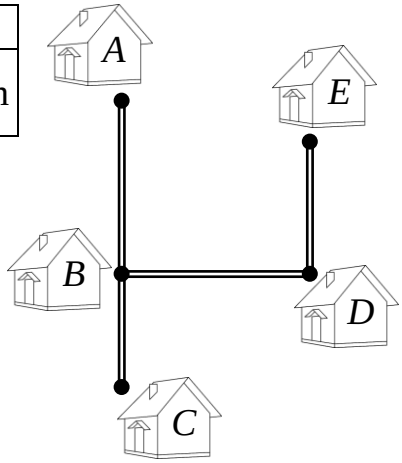
1st	2nd	3rd	4th

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

4. As shown in the diagram below, 5 persons (*A*, *B*, *C*, *D* and *E*) live in one village. The table below shows the sum of the distance travelled by the other 4 people when going to a one house. If a party is to be held in E's house, what is the total distance travelled, in meters, of the 4 people (*A*, *B*, *C* and *D*)?

Party house	*A*	*B*	*C*	*D*
Total distance travelled by 4 people	4180 m	2530 m	3610 m	3130 m



Answer : _____ m

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

5. Three motorcyclists started their rides at the same time. During the whole journey, they travelled for some time, took a rest for some time and then travelled to return to their homes at the same time. It's known that the first motorcyclist travelled twice as much as the second motorcyclist rested, the second motorcyclist travelled three times as much as the third motorcyclist rested and the third motorcyclist travelled four times as much as the first motorcyclist rested. If the sum of the speeds of the first and the third is 48 km/h, find the speed of the second one.

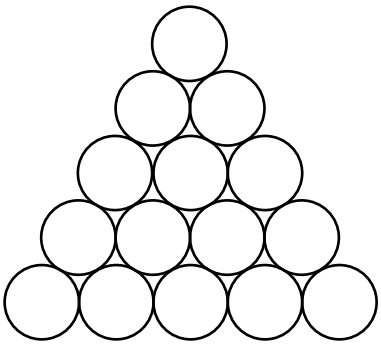
Note: Each motorcyclist travelled at the same speed for the whole journey.

Answer： _____ km/h

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

6. A pair of circles are called “neighbors” if they touch each other. How many pairs of “neighbors” are there in the diagram below?

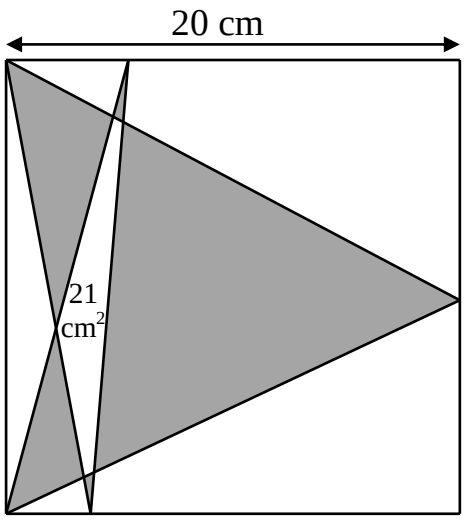


Answer : _____ pairs

ESSAY PROBLEMS

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7. In the diagram below, the square has side length 20 cm, and the area of the 'middle pentagon' inside the square is 21 cm^2. What is total area, in cm^2, of all the shaded regions?



Answer : _____ cm^2

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

8. In a chess tournament, each participant plays with every other participant exactly once. Each participant gets 1 point for a win, 0.5 point for a draw and 0 points for a loss. At most how many of the 40 participants can score 24 points or more?

Answer : _____ participants

ESSAY PROBLEMS

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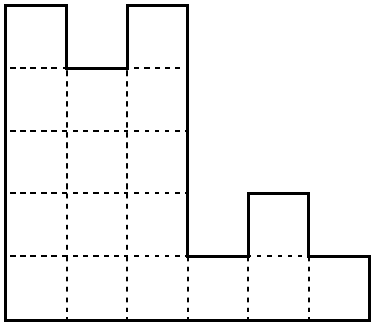
9. There are 8 consecutive non-zero positive integers such that their sum can be expressed as the sum of 7 consecutive positive integers and it cannot be expressed as the sum of 3 consecutive positive integers. What is the minimum value of the largest number among the 8 consecutive positive integers?

Answer : _____

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10. Dissect the diagram below into two congruent parts which may be rotated or reflected. Find the perimeter, in cm, of each part if the area of the original diagram is 72 cm^2.



Answer : _____ cm

ESSAY PROBLEMS

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11. Five teams took park in a soccer tournament, where each team plays each other exactly once. Each team gets 3 points for a win, 1 point for a draw and 0 points for a loss. After all matches has been played, is it possible that the total point scored of these five teams are five consecutive positive integers? If yes, provide an example.

example

v.s.	I	II	III	IV	V	points
I	■					
II		■				
III			■			
IV				■		
V					■	

☐ My answer is **YES**,

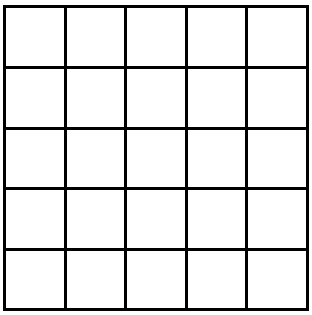
☐ My answer is **NO**, the reason is

Answer:

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

12. In each cell of the 5×5 grid shown below, Peter may place several stones or leave it empty such that the total number of stones for each and every row and columns are all different. What is the minimum number of stones that Peter must use to satisfy the conditions?

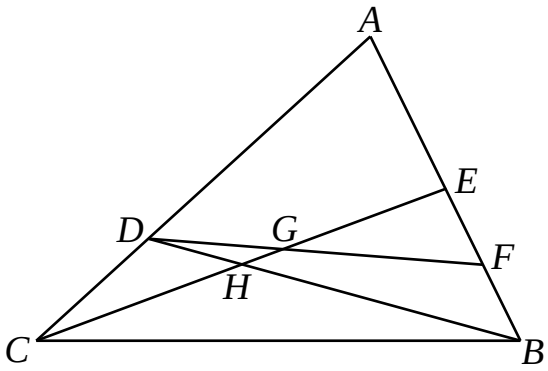


Answer: _____ stones

ESSAY PROBLEMS

Name: _____ Country: _____ Index Number: _____

13. In the diagram below, the area of triangle *EFG* is equal to the area of triangle *CDG*, the ratio between the area of triangles *EBC* and *BCD* is 3 : 2 and *AD* : *DC* = 2 : 1. If the area of triangle *BDF* be 29 cm^2, then what is the area, in cm^2, of triangle *ABC*?



Answer : _____ cm^2